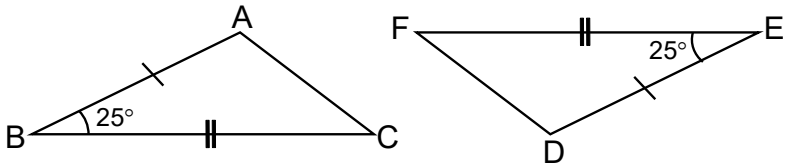
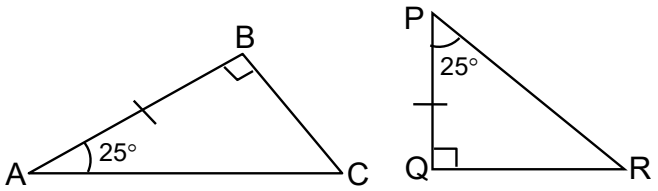
ASSIGNMENT

1. State the condition under which $\Delta ABC \cong \Delta DEF$ is the following case.



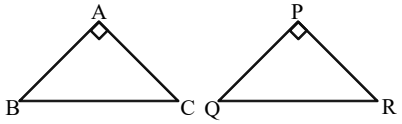
1) ASA 2) SAS 3) SSS 4) RHS

2. State the condition under which $\Delta ABC \cong \Delta PQR$ in the following case?



1) ASA 2) SAS 3) SSS 4) RHS

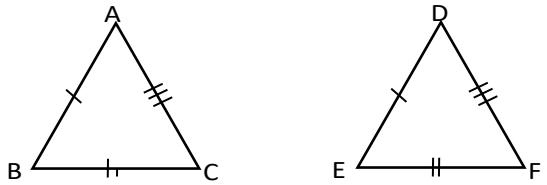
3. In the given figure, $\angle C$ is equal to $\angle R$ and BC = QR.



By which congruency criterion are ΔABC and ΔPQR congruent?

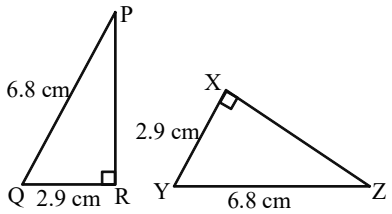
1) SSS 2) SAS 3) ASA 4) RHS

4. By which congruency criterion are ΔABC and ΔDEF congruent?
If AD = DE



1) SSS 2) SAS 3) ASA 4) RHS

5. The given figure shows two triangles.



Which of the following statements is correct?

11. A bird flies 6 km in 2 minutes. The expression which represents the distance covered by the bird in its flying time 't' minutes is ____

1) 6t 2) 3t 3) 4t 4) 6t+2

12. Which expression represents 5 less than thrice the square of a number?

1) $3n^2-5$ 2) $3n^2 + 5$ 3) $- 3n^2- 5$ 4) $3n^2+5$

13. Six more than seven times a number is thirty-four. Find the number.

1)7 2)4 3)5 4)21

14. The difference between three times a number and five is sixteen. Find the number.

1)7 2)4 3)5 4)21

15. A number decreased by half of the number is four. Find the number.

1)7 2)4 3)5 4)8

16. Four times the sum of twice a number and six is thirty-two. Find the number.

1)7 2)4 3)5 4)1

17. Two-thirds of the sum of three times a number and six is ten. What is the number?

1)7 2)3 3)5 4)1

18. The 22 students in Ms. Smith's 2nd grade class each have a sibling or a pet. If 14 students have a sibling and 18 students have a pet, how many students have both a sibling and a pet?

1)12 2)14 3)10 4)12

19. Jane spent 42 rupees for shoes. This was 14 rupees less than twice what she spent for a blouse. How much was the blouse?

1)28 2)24 3)20 4)26

20. A factor of $x^3 - 1$ is

1) $x - 1$ 2) $x^2 + x + 1$ 3) either A or B 4) none

21. If $(x - 2)$ is one factor of $x^2 + ax - 6 = 0$ and $x^2 - 9x + b = 0$, then $a + b =$

1) 15 2) 13 3) 11 4) 10

22. The remainder obtained when $t^6 + 3t^2 + 10$ is divided by $t^3 + 1$ is

1) $t^2 - 11$ 2) $t^3 - 1$ 3) $3t^2 + 11$ 4) none of these

23. If $x - \frac{1}{x} = \sqrt{6}$, then $x^2 + \frac{1}{x^2} =$

1) 2 2) 4 3) 6 4) 8

24. If $x^2 + \frac{1}{x^2} = 79$, then $x + \frac{1}{x} =$

1) $\sqrt{75}$ 2) 9 3)5 4)6

13. How can the expression $\frac{18ab^2 \times (3a)^3 \times (4b)^2}{12a^2 \times (ab^2)^2}$ be simplified?

1) $16ab^2$ 2) $24a^2b$ 3) 216 4) 648

14. The distance between the Sun and Jupiter is 77,80,00,000 km. What is the scientific notation (standard form) of this distance?

1) 7.78×10^8 2) 77.8×10^7 3) 0.778×10^9 4) 778×10^6

15. How can the number 90252 be expanded?

1) $9 \times 10^4 + 2 \times 10^2 + 5 \times 10^1 + 2 \times 10^0$

2) $9 \times 10^4 + 2 \times 10^2 + 5 \times 10^1 + 2 + 10^0$

3) $9 \times 10^3 + 10 \times 10^2 + 2 \times 10^1 + 5 \times 10^0 + 2 + 1$

4) $9 \times 10^4 + 10 \times 10^3 + 2 \times 10^2 + 5 \times 10^1 + 2 + 10^0$

16. What is the value of the expression {– (–3) × (–3) × (–3) × (–3) – (–3) × (–3) × (–3) × (–3) – (–3) × (–3) × (–3) × (–3)}?

1) -3^5 2) $(-3)^4$ 3) $(-3)^{12}$ 4) $12\ (-3)^3$

17. Which expression is equivalent to the expression $(3^2 + 3^3 + 3^4 + 3^5)$?

1) $3^3 \times 2 \times 5$ 2) $3^2 \times 2^2 \times 5$ 3) $3^2 \times 2^3 \times 5$ 4) $3^3 \times 2 \times 5^2$

18. Which is the greatest prime factor of $27^{10} - 243^5$?

1) 2 2) 3 3) 11 4) 13

19. The value of 'y' that satisfies the equation

$5^w \times 7^x \times 2^y \times 3^z = 36^3 \times 35^2 \times 30^2 \times 21$ is

1) 4 2) 5 3) 6 4) 8

20. The number of prime factors in

$[(6)^{12} \times (35)^{28} \times (15)^{16}]/[(3)^{16} \times (5)^{43} \times (14)^{12} \times (21)^{11} \times (7)^4]$ is

1) 15 2) 7 3) 5 4) 3

21. If abc = 1, then $\left(\frac{-1}{1+a+b^{-1}} - \frac{1}{1+b+c^{-1}} - \frac{1}{1+c+a^{-1}}\right) = ?$

1) ab 2) 0 3) $\frac{1}{ab}$ 4) 1

22. If x, y, z are real numbers, then the value of $\sqrt{x^{-2}y^2}.\sqrt{y^{-2}z^2}.\sqrt{z^{-2}x^2}$?

1) xyz 2) $\sqrt{xyz}$ 3) $\frac{1}{xyz}$ 4) 1

23. If $3^n = 729$, find the values of 3^{n-3}

1)27 2)81 3)243 4)9

ASSIGNMENT

1. If A : B = 3 : 2 and B : C = 5 : 4, then A : B : C is

   1) 15 : 10 : 7 2) 15 : 10 : 6 3) 15 : 10 : 4 4) 15 : 10 : 8

2. If you divide Rs 1160 between A, B and C. In such a way that A : B = 4 : 3 and B : C = 2 : 5, then the share of B is

   1) Rs 230 2) Rs 210 3) Rs 240 4) Rs 200

3. A, B and C invested Rs.17500, Rs.20000 and Rs.22500 in a business. If they got a profit of Rs.4800 at the end of the year, then the share of C is

   1) 1600/- 2) 1700/- 3) 1800/- 4) 2000/-

4. If $\frac{A}{3} = \frac{B}{4} = \frac{C}{5} = x$, where A + B + C = 3600, then the value of x is

   1) 400 2) 300 3) 500 4) 200

5. The ratio of red balls, green balls and yellow balls in a bag is 5 : 3 : 2. If the number of yellow balls is 12, then the total number of balls is

   1) 70 2) 80 3) 60 4) 90

6. If the ratio of angles of a scalene triangle is 3 : 4 : 8, then the largest angle is

   1) 98^0 2) 95^0 3) 96^0 4) 94^0

7. If A gets 5 times as much as B and B gets 4 times as much as C, then if you divide as 1000/- the share A and B are

   1) 800/-, 170/- 2) 800/-, 140/- 3) 800/-, 160/- 4) 800/-. 150/

8. In a cash box the ratio of number of Rs. 10 notes and Rs. 20 notes and Rs. 50 notes are in the 5 : 3 : 2. If the total number of notes is 150 the value of 10 rupees notes is

   1) 760/- 2) 740/- 3) 750/- D 720/-

9. If x : y = $2\frac{1}{3} : 3\frac{1}{4}$ and y : z = $5\frac{1}{2} : 2\frac{1}{4}$, then x : z is

   1) $\frac{77}{6} : \frac{117}{16}$ 2) $\frac{78}{6} : \frac{113}{16}$ 3) $\frac{74}{6} : \frac{117}{16}$ 4) $\frac{77}{6} : \frac{115}{16}$

10. If A + 2B + 3C = 510 where A = 3B, C = 4B, then the value of C is

    1) 110 2) 120 3) 130 4) 100-

11. If A = 15 : 45 = 40 : 120, B = 33 : 121 = 9 : 96, C = 24 : 28 = 36 : 48 and D = 8 : 6 = 4 : 5, then which one of the following is in proportion?

    1) B 2) A 3) D 4) C

12. If 42 : 56 = 51 : x, then the value of x is

    1) 85 2) 34 3) 68 4) 136

1) $2:\sqrt{3}$ 2) $1:\sqrt{3}$ 3) $\sqrt{3}:1$ 4) $4:\sqrt{3}$

50. Length and Breadth of a rectangle is 7 m and 3.5 m respectively. Area of gratest circle inscribed in a rectangle is

1) 9.625 2) 9.74 3) 9.8 4) 9.725

51. The length of a rectangle is twice its breadth. If its length is decreased by 5 cm and breadth is increased by 5 cm, the area of the rectangle is increased by 75 sq.cm. The length of the rectangle is

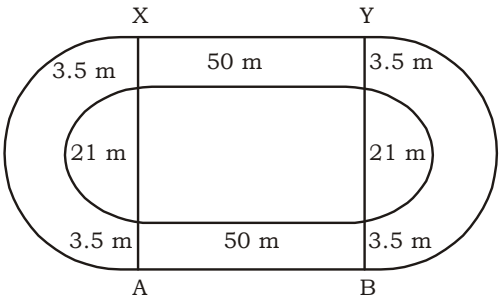
1) 40 cm 2) 15 cm 3) 20 cm 4) 18 cm

52. If each side of a square is increased by 25%, find the percentage change in its area.

1) 62.25 2) 59.52 3) 58.75 4) 56.25

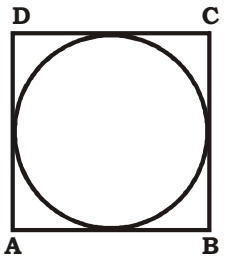
53. A room is half as long as it is broad. The cost of carpeting the at Rs.5 per sq.m is Rs.270 and the cost of papering the four walls at Rs.10 per m^2 is Rs.1420. If a door and 2 windows occupy 8 sq.m, find the height of the room.

1) 4 m 2) 4.5 m 3) 5 m 4) 5.5 m



1) 178 m 2) 198 m 3) 188 m 4) 168 m

54. If the side of the square is 14 cm, then ratio of their perimeters is



1) 11 : 14 2) 11 : 15 3) 14 :11 4) 15 : 11

ASSIGNMENT KEY

1. 1	2. 3	3. 3	4. 1	5. 2	6. 3	7. 3	8. 3	9. 3
10. 4	11. 1	12. 2	13. 3	14. 1	15. 2	16. 2	17. 1	18. 1
19. 4	20. 2	21. 2	22. 1	23. 3	24. 1	25. 2	26. 4	27. 3
28. 2	29. 4	30. 2	31. 1	32. 3	33. 1	34. 3	35. 1	36. 2
37. 3	38. 3	39. 3	40. 3	41. 2	42. 3	43. 2	44. 3	45. 4
46. 4	47. 4	48. 2	49. 4	50. 1	51. 1	52. 4	53. 3	54. 3

37. If 2.4 (3 – x) – 0.6 (2x – 3) = 0, then the value of x is

1) $\frac{5}{4}$ 2) $\frac{5}{6}$ 3) $\frac{5}{2}$ 4) $\frac{5}{8}$

38. If $\frac{2x+14}{3x+6} = 4$, then the value of x is

1) –2 2) –3 3) –1 4) –4

39. If $\frac{0.25+x}{3} = x + 0.5$, then the value of x is

1) $-\frac{3}{8}$ 2) $-\frac{5}{8}$ 3) $-\frac{1}{8}$ 4) $-\frac{7}{8}$

40. If $\frac{3}{4}(7x-1) - x = 2x - \frac{1-x}{2} + \frac{3}{2}$, then the value of x is

1) – 2 2) 3 3) 1 4) – 4

41. If five sixths a number exceeds it by two thirds of the number by 27, then the number is

1) 168 2) 162 3) 164 4) 166

42. If a number is as much greater than 22 as is less than 48, then the number is

1) 36 2) 37 3) 35 4) 38

43. If one third of a number is increased by 10 and the result is multiplied by 2. If the final result is 30, then the number is

1) 16 2) 15 3) 17 4) 18

44. If one of the two numbers exceeds the other by 4, If two times the larger number is added to 3 times the smaller gives 83, then the numbers are

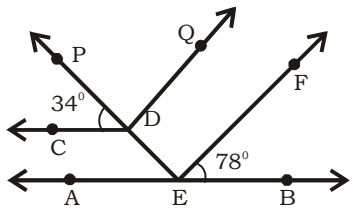
1) 15, 19 2) 14, 18 3) 13, 17 4) 16, 20

45. If 10 is added to 4 times a number the result is 5 less than five times the number, then the number is

1) 16 2) 17 3) 15 4) 18

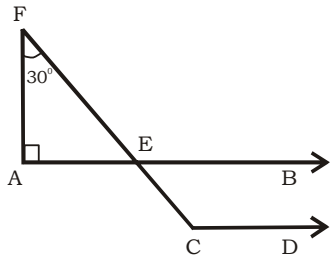
SELF TEST KEY

1.	**3**	**2.**	**1**	**3.**	**3**	**4.**	**1**	**5.**	**2**
6.	**4**	**7.**	**3**	**8.**	**2**	**9.**	**3**	**10.**	**2**
11.	**4**	**12.**	**3**	**13.**	**1**	**14.**	**2**	**15.**	**1**
16.	**3**	**17.**	**3**	**18.**	**1**	**19.**	**1**	**20.**	**1**
21.	**2**	**22.**	**4**	**23.**	**1**	**24.**	**2**	**25.**	**2**
26.	**2**	**27.**	**3**	**28.**	**1**	**29.**	**3**	**30.**	**2**
31.	**3**	**32.**	**3**	**33.**	**3**	**34.**	**2**	**35.**	**2**
36.	**1**	**37.**	**3**	**38.**	**3**	**39.**	**2**	**40.**	**3**
41.	**2**	**42.**	**3**	**43.**	**2**	**44.**	**1**	**45.**	**3**



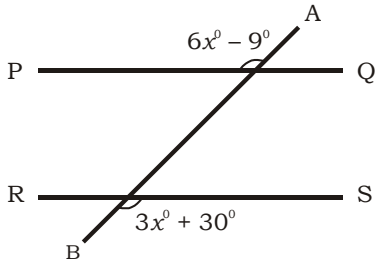
1) 112^0 2) 146^0 3) 102^0 4) 68^0

19. In the figure given below, AB || CD and $\angle F = 30^0$. Then $\angle ECD$ is



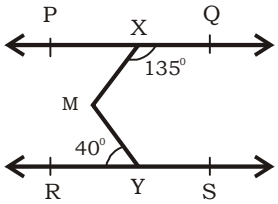
1) 130^0 2) 120^0 3) 150^0 4) 60^0

20. The value of x^0 that will make PQ parallel to RS is



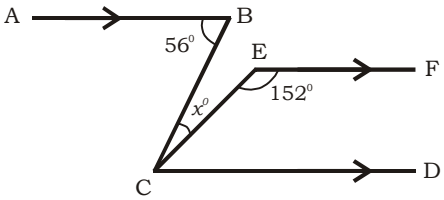
1) 17^0 2) 50^0 3) 13^0 4) 27^0

21. In the figure given below, PQ || RS, $\angle MXQ = 135^0$ and $\angle MYR = 40^0$. Then $\angle XMY$ is



1) 45^0 2) 85^0 3) 120^0 4) 40^0

22. In the figure given below, AB || CD || EF. Then the value of x is



1) 28^0 2) 72^0 3) 56^0 4) 80^0

23. In the figure, BC||DE, then $\angle ADE$ is

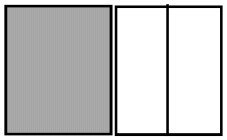
10. x = $\sqrt{0.5}$, y =0.5 then relation between x and y

1)x = y 2)x < y 3)x > y 4) None of these

11. How would you write the fraction $\frac{2}{100}$ as a decimal?

1) 0.2 2)0.02 3)0.002 4)0.0002

12. What decimal represents the shaded part?



1)0.25 2)0.40 3)0.50 4)0.75

13. Arrange the numbers 0.02, 0.2 , 0.02, 0.202 in decending order

1)0.202, 0.2, 0.02, 0.002 2)0.02, 0.2,0.02,0.002

3)0.202, 0.02,0.2,0.002 4)0.2, 0.02,0.02,0.002

14. $\sqrt{(0.04)^2+(0.03)^2}$ =

1)0.0005 2)0.025 3)0.5 4)0.05

15. The value of $\sqrt{2980}$ correct to two decimal place is:

1) 17.26 2) 54.59 3) 58.75 4) 74.29

16. Given that $(5.142)^2$ = 26.44, find $\sqrt{264\,400}$.

1) 5 142 2) 514.2 3) 51.42 4) 5.142

17. Which of the following values is not equal to 5?

1) $\sqrt{5^2}$ 2) $\left(\sqrt{5}\right)^2$ 3) $\sqrt{(-5)^2}$ 4) $-\sqrt{(5)^2}$

18. Given that $\sqrt{45}$ = 6.708 and $\sqrt{4.5}$ = 2.121, find the value of $\sqrt{4,500}$ + $\sqrt{450}$.

1) 882.9 2) 458.7 3) 88.29 4) 45.87

19. Given that $\sqrt{64.52}$ = 8.032, what is the value of $(80.32)^2$?

1) 64,520 2) 6,452 3) 645.2 4) 64.52

20. Given that $\sqrt{6.4}$ = 2.53, find the value of $\sqrt{640}+\sqrt{64}$.

1) 33.30 2) 27.83 3) 10.53 4) 3.33

21. Given that $(3.9268)^2$ = 15.42, find the value of $\sqrt{154\,200}$.

1) 3926.8 2) 392.68 3) 39.268 4) 3.9268

FRACTIONS & DECIMALS

FRACTIONS AND THEIR CLASSIFICATION, EQUIVALENT FRACTIONS

Fundamental Concepts of Fractions

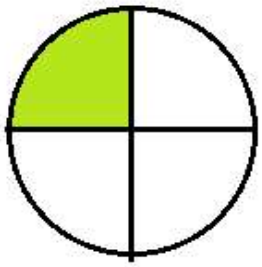
Introduction to Fractions

Fraction :A fraction (from Latin: fractus means "broken") represents a part of a whole or, more generally, any number of equal parts.

Numerator : The top number of a fraction is called its numerator.

Denominator: The bottom part of a number is its denominator.

A fraction is the number of shaded parts divided by the number of equal parts as shown below:



Number of shaded parts in numerator

Number of equal parts in denominator

Looking at the above figure

There are four equal parts, giving a denominator of 4. One of the parts is shaded, giving a numerator of 1.

Hence fraction is denoted by $\frac{1}{4}$

Unit fraction :A unit fraction is a vulgar fraction with numerator equal to 1.

Eg. : $\frac{1}{8}$.

Proper fraction: When the numerator and the denominator are both positive,the fraction is called proper if the numerator is less than the denominator.

Improper fraction: When the numerator and the denominator are

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CLASS VII

FOUNDATION MATHEMATICS

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